GenePlexus: A web server and Python package for gene discovery using network-based machine learning

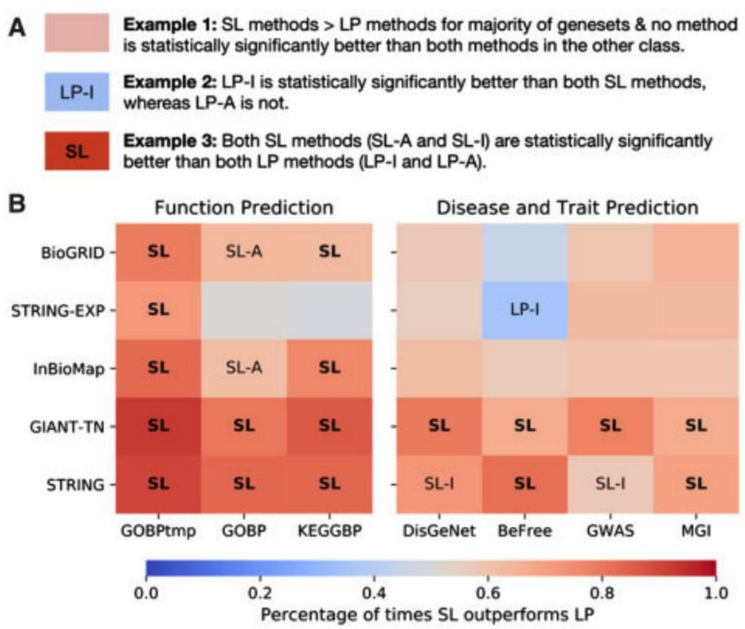
Christopher A Mancuso[1], Renming Liu[1], Patrick S. Bills[2], Douglas Krum[2], Jacob Newsted[2], Arjun Krishnan[1,3]

[1]Department of Computational Mathematics, Science and Engineering, Michigan State University, East Lansing, MI, 48824
[2]Data Management and Analytics, IT Services, Michigan State University, East Lansing, MI, 48824
[3]Department of Biochemistry and Molecular Biology, Michigan State University, East Lansing, MI, 48824

CMSE MICHIGAN STATE UNIVERSITY

GenePlexus Method

- An important task in biology is computational predicting pathways, traits, and diseases that individual genes are involved in (a task referred to as gene classification).
- Large-scale gene interaction networks are key components of modern gene classification approaches.
- We have shown that using network connections as features in supervised machine learning models outperforms the widely used method of label propagation. https://doi.org/10.1093/bioinformatics/btaa150



We compare four network-based methods

- Supervised Machine Learning (SL) (l2 regularized log reg)
  - SL-A: feature vectors are rows of adjacency matrix
  - SL-I: feature vectors are rows of influence matrix
  - SL-E: feature vectors node embeddings from node2vec
- Label Propagation (LP)
  - LP-I: influence matrix as transformation operator
  - LP-A: adjacency as transformation operator

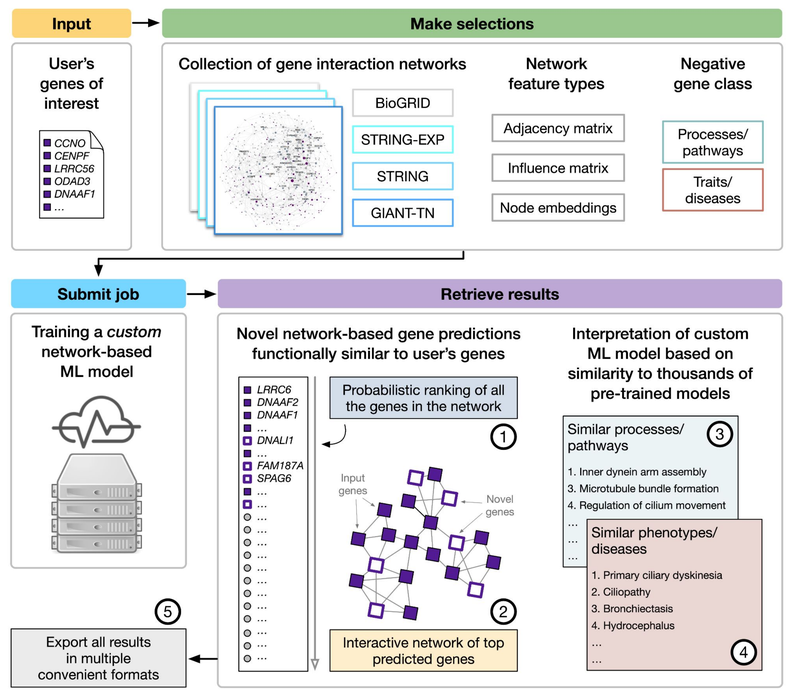
* Influence matrix created by random walk with restart kernel

We compared methods across many tasks (function, disease, and trait prediction) consisting of 1,000s of individual genesets.

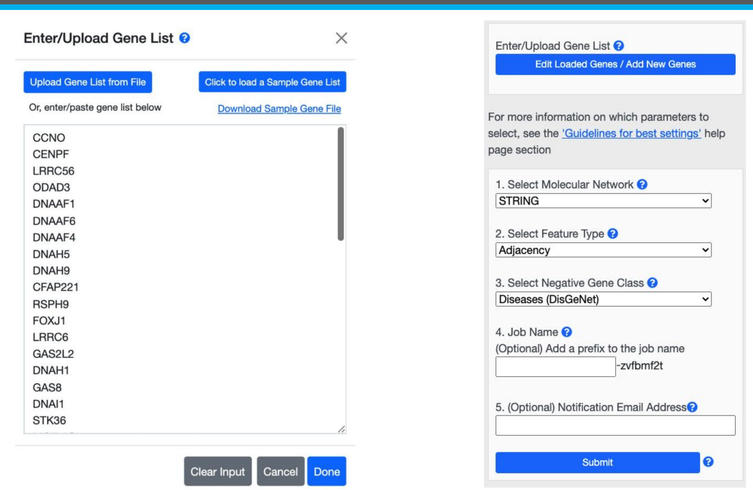
SL is almost always the best performing method.

Web Server Overview

GenePlexus Webserver : www.geneplexus.net



Inputs

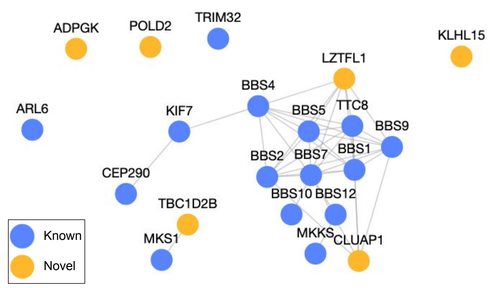


- A custom supervised machine learning model is trained using the user supplied genes as positive class examples.
- A user can select 1) the most suitable network, 2) how the network is represented in the model, and 3) how negative genes are selected

Genome-wide Predictions

Rank	Entrez	Symbol	Name	Probability	Known/Novel	Training-Label
1	583	BBS2	Bardet-Biedl syndrome 2	0.99	Known	P
2	54585	LZTFL1	leucine zipper transcription factor like 1	0.99	Novel	U
3	55212	BBS7	Bardet-Biedl syndrome 7	0.99	Known	P
4	129880	BBS5	Bardet-Biedl syndrome 5	0.99	Known	P

Network Visualization



(Above) Prediction of how related all genes in network are to the input set

(Left) Interactive visualization of the connectivity of the top predicted genes

Model Interpretability

The learned coefficients from the model trained on the user supplied genes are compared to the coefficients from thousands of models trained using gene sets from known biological processes, pathways and diseases.

Note: this is distinct from gene set enrichment which is what other softwares provide

Rank	ID	Name	Similarity
1	GO:0070286	axonemal dynein complex assembly	21.97
2	GO:0036159	inner dynein arm assembly	21.63
3	GO:0036158	outer dynein arm assembly	21.58
4	GO:0001578	microtubule bundle formation	18.81
5	GO:0035082	axoneme assembly	18.02
6	GO:0003352	regulation of cilium movement	17.79
7	GO:0060632	regulation of microtubule-based movement	15.00
8	GO:0006858	extracellular transport	13.38
9	GO:0060295	regulation of cilium movement involved in cell motility	13.05
9	GO:1902019	regulation of cilium-dependent cell motility	13.05

Contact and References

- mancus16@msu.edu
- https://doi.org/10.1093/nar/gkac335
- NIH R35 GM128765/F32 F32GM134595

Links to all papers and software